Why polar energy shaping helps

From a learned correction to a testable geometric explanation

10 October 2026 — Job 034 follow-up; companion to Report 7

Result to explain. With unchanged full-alphabet D2, learned symbol energies bring ODMA–polar close to fixed dense codebooks. Full-grid mean PUPE at $B = 12/14$ is 0.17116/0.28776, versus fixed polar 0.18346/0.29620 and dense 0.17290/0.28526. These are three-seed, distinct-message means, not an equivalence proof. We derive the balancing correction, identify the structure it cannot remove, and distinguish a geometric certificate from a decoding claim. No new receiver or production training is introduced here.

1 The encoder and its exact second moment

Split $w = (p, v)$ into B_p placement bits and B_c coded-payload bits. Put $P = 2^{B_p}$, $V = 2^{B_c}$ and $M = PV$. The linear CRC/polar generator $L \in \mathrm{GF}(2)^{N_c \times B_c}$ produces the sign vector

$$b_i(v) = (-1)^{\ell_i^T v}, \qquad \phi(p, v) = S_p D_p b(v), \qquad D_p = \mathrm{diag}(\sqrt{e_p}), \quad \sum_i e_{p,i} = 1. \tag{1}$$

Here ℓ_i^T is row i of L; $S_p \in \mathbb{R}^{n \times N_c}$ inserts symbols at distinct positions $r(p, i)$. Thus $S_p^T S_p = I$, and every waveform has energy one. The learnable bank e has shape $P \times N_c$, not $N_c \times M$. The current implementation uses $e_p = \mathrm{softmax}(\theta_p)$, so energies are positive. Polar bits and placements remain fixed. The explicit reference is

$$\Phi = [S_1 D_1 B_{\mathrm{pol}} \ \cdots \ S_P D_P B_{\mathrm{pol}}] \in \mathbb{R}^{n \times M}, \qquad B_{\mathrm{pol}} = [b(v)]_v \in \mathbb{R}^{N_c \times V}, \qquad y = \Phi a^\star + z. \tag{2}$$

The count vector $a^\star$ sums to the known activity K; the real noise is $z \sim \mathcal{N}(0, \sigma^2 I)$.

For a uniform payload, a nonzero binary linear function is zero or one equally often. Consequently,

$$Q_{ij} := \mathbb{E}_v[b_i(v) b_j(v)] = \mathbb{E}_v[(-1)^{(\ell_i \oplus \ell_j)^T v}] = \begin{cases} 1, & \ell_i = \ell_j, \\ 0, & \ell_i \neq \ell_j. \end{cases} \tag{3}$$

So repeated generator rows mean *identical symbol functions*, not merely correlated samples. We can compute the whole codebook's normalized frame operator without enumerating payloads:

$$\boxed{C := \frac{1}{M} \Phi \Phi^T = \frac{1}{P} \sum_p S_p D_p Q D_p S_p^T}, \qquad \mathrm{tr}\, C = 1. \tag{4}$$

C is an uncentered second moment, not generally a covariance. Its diagonal is particularly simple:

$$C_{rr} = \frac{q_r}{P}, \qquad q_r = \sum_{p,i:\, r(p,i)=r} e_{p,i}. \tag{5}$$

Unit energy fixes each *column* of Φ; balanced q_r fixes total energy across its *rows*. They are different conditions. Current sizes are $n = 256, N_c = 32, B_c = 6$, with $P = 64/256$ at $B = 12/14$.

Computation and checks. After forming L, Q, moments cost $O(PN_c^2 + n^2)$ with no 2^{B_c} table; balancing costs $O(PN_c + n)$ per iteration. The pattern bank still scales as 2^{B_p}, so keep B_p bounded when discussing large B. Encoding costs $O(N_c \log N_c + n)$ with dense waveform output; D2 still keeps 2^B states. `benchmarks/polar_energy_geometry.py` computes the diagnostics. Regression tests verify explicit-matrix agreement, the analytic gradient, balancing constraints and the known-K covariance identity below.

2 Why balancing is relevant to D2

Let $G = \Phi^T\Phi$. For unit columns, D2's scalar statistic has the exact decomposition

$$t_j = \phi_j^T(y - \Phi\widehat{a}) + \widehat{a}_j = a_j^\star + \sum_{k\neq j} G_{jk}(a_k^\star - \widehat{a}_k) + \phi_j^T z. \tag{6}$$

If the other errors were zero-mean, uncorrelated and of common variance ν, the interference variance would be $\nu\sum_{k\neq j} G_{jk}^2$. D2 approximates this with the mean posterior variance; learned scaling and damping then adjust it. Exact-K dependence, unequal uncertainties and observation reuse prevent this from being exact inference.

Nevertheless, its average geometric interference factor has an exact identity:

$$\frac{1}{M}\sum_j\sum_{k\neq j} G_{jk}^2 = \frac{\|\Phi^T\Phi\|_F^2 - M}{M} = M\,\mathrm{tr}(C^2) - 1 \tag{7}$$

$$= \frac{M}{n} - 1 + M\left\|C - \frac{I}{n}\right\|_F^2. \tag{8}$$

Since $\mathrm{tr}\,C = 1$, the lower bound is $M/n - 1$, attained exactly when $C = I/n$. This is the unit-norm tight-frame criterion underlying the classical frame potential [1]. It is an optimum of *average squared correlation*, not of PUPE. All remaining questions depend on that distinction.

For interpretation, use the dimensionless defect

$$\boxed{\Delta := n\left\|C - \frac{I}{n}\right\|_F^2 = \underbrace{n\sum_r (C_{rr} - 1/n)^2}_{\Delta_{\mathrm{row}}} + \underbrace{n\sum_{r\neq s} C_{rs}^2}_{\Delta_{\mathrm{off}}}.} \tag{9}$$

The first term is exactly the squared coefficient of variation of row energies. The second records correlations between physical coordinates. Balancing removes the first term; it does not automatically remove the second. The participation measure used in the result review is $1/\,\mathrm{tr}(C^2) = n/(1+\Delta)$, not the literal matrix rank.

A precise conditional equivalence. If every row of L is distinct, then $Q = I$ and C is diagonal. If the support admits balanced positive energies, setting $q_r = P/n$ therefore gives $C = I/n$ exactly. The structured encoder then attains the unrestricted tight-frame minimum in (8). This is a useful formal statement, but it neither makes its Gram matrix equal to a dense codebook's nor makes decoding equivalent.

Our current code does not satisfy that premise. The effective 32×6 generator has 29 distinct rows, including one zero row. For $r \neq s$, (4) gives

$$C_{rs} = \frac{1}{P}\sum_{p,i\neq j:\, r(p,i)=r,\, r(p,j)=s} \sqrt{e_{p,i}e_{p,j}}\,\mathbf{1}\{\ell_i = \ell_j\} \;\geq 0. \tag{10}$$

With strictly positive energies, a repeated row creates a positive off-diagonal contribution which other patterns cannot cancel in this family. Thus exact $C = I/n$ is impossible here at finite logits. Allowing some energies to vanish can remove individual contributions but changes the effective code and may damage distances; it is not a free fix. This is a statement about the selected small-payload construction, not about all polar codes or large B.

3 Deriving the deterministic correction

Treat each allowed placement (p, i) as an edge joining pattern p to resource $r(p, i)$. We want

$$e_{p,i} \geq 0, \qquad \sum_i e_{p,i} = 1, \qquad \sum_{p,i:\, r(p,i)=r} e_{p,i} = \frac{P}{n}. \tag{11}$$

These are linear constraints: unit mass per pattern and equal total mass per channel coordinate. Among all feasible allocations, choose the one nearest the original uniform profile $e^0_{p,i} = 1/N_c$ in generalized KL divergence:

$$\min_{e \text{ satisfying (11)}} \sum_{p,i} \left[e_{p,i} \log \frac{e_{p,i}}{e^0_{p,i}} - e_{p,i} + e^0_{p,i} \right]. \tag{12}$$

This is a strictly convex problem on a convex feasible set, so its energy solution is unique when feasible. For an interior solution the Lagrangian derivative is

$$\log(e_{p,i}/e^0_{p,i}) + \alpha_p + \beta_{r(p,i)} = 0, \qquad e_{p,i} = e^0_{p,i}\, u_p v_{r(p,i)}, \tag{13}$$

with positive scaling factors u, v. This motivates alternating two exact marginal corrections:

$$e_{p,i} \;\leftarrow\; e_{p,i} \frac{P/n}{q_{r(p,i)}}, \qquad e_{p,i} \;\leftarrow\; \frac{e_{p,i}}{\sum_j e_{p,j}}. \tag{14}$$

This is iterative proportional fitting, or matrix scaling on the allowed edges [2]. Starting at e^0, it preserves the multiplicative form (13). A positive converged solution satisfies both the constraints and the stationarity equations, hence solves (12). With a strictly positive feasible allocation, the standard alternating-projection result applies. Boundary solutions need separate care.

Feasibility is not automatic. Every resource must be covered, but coverage is insufficient. For nonnegative allocations, a necessary and sufficient transport condition is

$$|\mathcal{P}| \leq \frac{P}{n} |N(\mathcal{P})| \quad \text{for every subset of patterns } \mathcal{P}, \tag{15}$$

where $N(\mathcal{P})$ is their set of reachable resources. It follows from the source-pattern-resource-sink max-flow cut condition. Strict positivity on every allowed edge is an additional requirement. Our implementation stops only after checking both marginals and reports nonconvergence rather than silently labelling an allocation balanced. All six current banks converge to positive solutions.

3.1 Going beyond balancing without guessing another encoder family

The natural next analytic objective is the full defect $\mathcal{J} = \|C - I/n\|_F^2$, not only its diagonal part. For $a_p = \sqrt{e_p}$ and $H_p = S_p^T (C - I/n) S_p$, direct differentiation of (4) gives

$$\frac{\partial \mathcal{J}}{\partial e_{p,i}} = \frac{2}{P\sqrt{e_{p,i}}} \sum_j (H_p)_{ij} Q_{ij} \sqrt{e_{p,j}}, \qquad \frac{\partial \mathcal{J}}{\partial \theta_{p,i}} = e_{p,i} \left(g_{p,i} - \sum_j e_{p,j} g_{p,j} \right), \tag{16}$$

where $g = \partial \mathcal{J}/\partial e$. The gradient has been checked against automatic differentiation. It gives a decoder-independent diagnostic or candidate initialization/loss. Unlike (12), this general objective is not claimed convex, nor is minimizing it claimed to minimize PUPE. No new training arm is implemented.

4 What remains after balancing: measured and unmeasured

The following moments are evaluated exactly from the six returned energy checkpoints and original codebooks, using (4); each entry averages three seeds. The balanced construction uses no received data or trained energies. These are new geometry diagnostics, not new decoding experiments.

B	Construction	Δ_{row}	Δ_{off}	Δ
12	Original uniform	0.102865	0.024089	0.126953
	Deterministic balance	0	0.023249	0.023249
	Learned energies	0.003583	0.014004	0.017586
	Fixed dense	0.000526	0.061400	0.061926
14	Original uniform	0.026324	0.006607	0.032931
	Deterministic balance	0	0.006567	0.006567
	Learned energies	0.002477	0.005926	0.008402
	Fixed dense	0.000134	0.015481	0.015615

At $B = 12$, learning both nearly balances rows and reduces the off-diagonal term by about 40% relative to balancing alone. This identifies a correction beyond inverse resource degree: energy is moved away from harmful repeated-symbol interactions. It establishes what changed, not how much of the PUPE gain each term caused. At $B = 14$, learning reduces the off-diagonal term by about 10%, but its residual row imbalance makes total Δ larger than deterministic balancing's. Thus the learned objective is not simply minimizing frame potential. More decisively, at $B = 14$ both balanced and learned polar have smaller defect than the *actual fixed dense* reference, despite dense's lower measured PUPE. Reducing average interference cannot by itself explain or guarantee closing that remaining gap. The earlier small-grid receiver swaps support an encoder benefit, but are not a full-grid causal decomposition.

4.1 Three reasons that a residual performance gap can survive

1. The codeword mean and activity law. Let $m_i = \mathbf{1}\{\ell_i = 0\}$. Then

$$\mu = \mathbb{E}_w \phi(w) = \frac{1}{P} \sum_p S_p D_p m, \qquad \Sigma = C - \mu\mu^T. \tag{17}$$

For IID uniform messages the noiseless sum has covariance $K\Sigma$; for K uniformly selected distinct messages it is $K(M-K)/(M-1)\,\Sigma$. These identities are exact. A nonzero μ is not random information; its common contribution $K\mu$ can be subtracted when K is known. That does not recover its spent transmission energy. Iterative D2 error covariance is different again, so neither C nor Σ alone gives its true conditional uncertainty.

2. Rare correlations and competing message sets. Frame potential averages squared correlations; it can hide a damaging tail. For two allowed count vectors a, a', the actual separation is

$$d^2(a, a') = \|\Phi(a - a')\|^2 = (a - a')^T G (a - a'). \tag{18}$$

Under AWGN, the binary equal-prior comparison error is $\mathcal{Q}(d/(2\sigma))$. Here $\mathcal{Q}$ is the Gaussian tail, unrelated to the symbol matrix Q. The multiplicity of close competing sets matters too. Multiuser aliases, within-pattern weighted distances and rare cross-pattern correlations are therefore separate diagnostics.

3. The receiver and its optimization. D2 replaces correlated, non-Gaussian uncertainty by calibrated scalar channels. Its loss and early stopping do not optimize PUPE directly. A remaining gap can be waveform geometry, decoder approximation or training; it cannot be assigned to the encoder solely from the displayed defect. Frozen-calibration swaps are informative but not optimal decoding.

5 Equivalence and next decisions

Geometric optimality is achievable as a theorem. Equation (8) proves an average-interference minimum; distinct symbol functions plus feasible balancing attain it. For the current repeated-symbol code, quantify the constrained residual instead. These are precise statements worth pursuing, even if no PUPE theorem follows.

Tightness alone cannot prove equal recovery. Consider two unit-column 2×4 matrices:

$$\Phi_1 = [e_1, e_1, e_2, e_2], \qquad \Phi_2 = [e_1, e_2, (e_1 + e_2)/\sqrt{2}, (e_1 - e_2)/\sqrt{2}]. \tag{19}$$

Both satisfy $\Phi\Phi^T = 2I$. Yet Φ_1 aliases messages even with one noiseless user, while Φ_2 has four distinct columns. Matching frame operators is much weaker than matching communication systems.

A sufficient exact system equivalence. If $\Phi_2 = O\Phi_1\Pi$ for an orthogonal channel rotation O and a message permutation Π, white noise and exchangeable message priors give the same statistical problem after rotating observations and relabelling messages. Equivalently, matching full Gram matrices up to that permutation is sufficient for such an isometry on the spans. Bayes performance then agrees; D2 with matched parameters is also equivariant to these transformations. No such relationship to the dense reference has been demonstrated.

An empirical equivalence claim needs its own design. Predefine an acceptable absolute PUPE margin and operating region, then use independent seeds and paired trials to bound the performance difference. A mean close to dense, or an interval containing zero, is not an equivalence test. Receiver inference and approximation remain fixed in that claim; it is not global encoder optimality.

5.1 Two active research directions

1. **Encoder geometry.** Compare uniform, deterministic-balanced and learned energies with matched receivers. Diagnose the residual before adding a geometry loss or encoder freedom. A simple deterministic construction matching learned PUPE would be a positive result.
2. **Tractable polar-aware inference.** Use Report 7 to replace exponential message state, measuring loss against full D2 on identical waveforms. Unknown activity, pattern association, same-pattern collisions and joint interference remain open. Keep known-K and unknown-K results separate.

The shared mixer is retired. Hash/coordinate-amplitude and sectioned-tree routes are parked as historical evidence, not declared mathematically inferior. Polar energy has the stronger current performance/structure case, but no completed scalable multiuser decoder yet. Broad new encoder families are unnecessary unless a diagnosed residual points to a specific missing freedom.

Scope of a possible conclusion. "Simple structured encoding plus geometric optimization matches the dense reference under the tested receiver and conditions" would be well supported by a successful controlled test. "No subtle code structure can outperform well-behaved dense encoding" is much stronger and does not follow from second moments, D2 comparisons or these finite experiments. Dense is a useful reference, not a proven optimum.

References

[1] D. G. Mixon, T. Needham, C. Shonkwiler and S. Villar, *Three proofs of the Benedetto–Fickus theorem*, 2021/2023. arXiv:2112.02916. Background for the frame-potential criterion, not a URA theorem.

[2] I. Csiszár, *I-divergence geometry of probability distributions and minimization problems*, Annals of Probability 3(1), 146–158, 1975. Author paper. Background for KL projection and iterative fitting.